

AI-Based Restoration of Degraded Images for Semiconductor Inspection

KLA Hackathon 2026 – SEMICON India | Solution Submission

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- * **Team Architecture:** Nonlinear Activation Free Super-Resolution (NAFNet-SR)
- * **Task Objective:** End-to-end physics-informed restoration mapping 128x128 noisy degraded inputs to 256x256 ground truth wafer patterns.
- * **Key Metric Results:** Peak Validation PSNR: 28.16 ± 5.02 dB (vs 23.01 ± 3.65 dB Bicubic baseline), SSIM: 0.7661 ± 0.1571 .
- * **Target Hardware:** NVIDIA GeForce RTX 3050 Laptop GPU (Ampere, 4GB VRAM) | Mixed Precision (TF32)

1. Problem Understanding & Task Formulation

Physical Inspection Constraints in Modern Semiconductor Wafer Fabrication

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- * **Semiconductor Inspection Challenge:** Physical optical and e-beam imaging systems suffer from low photon counts, high speckle noise, thermal sensor noise, and limited dynamic range.
- * **Restoration Objective:** Transform 128x128 single-channel noisy degraded inputs (NoisyLR) to 256x256 clean high-resolution ground truth (GT) wafer images.
- * **Core Invariant 1:** Single-channel float32 grayscale format with physical reflectivity dynamics.
- * **Core Invariant 2:** NoisyLR pixel values extend outside $[0, 1]$ due to speckle noise -> Strictly unclipped at dataloader.
- * **Core Invariant 3:** Model output MUST be strictly clamped to $[0.0, 1.0]$.

2. Dataset Analysis & Degradation Characteristics

Multi-Modal Noise Dynamics & Dynamic Range Properties

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- * **1. Multiplicative Speckle Noise:** $I_{\text{noisy}} = I + I * N(0, \sigma_s^2)$ creates signal-dependent fluctuations in bright wafer traces.
- * **2. Additive Gaussian Thermal Noise:** $I_{\text{noisy}} = I + N(0, \sigma_g^2)$ perturbs dark substrate regions, pushing raw float32 pixels outside $[0, 1]$.
- * **3. Spatial Downsampling:** 2x resolution decimation causing high-frequency edge blur and line aliasing.
- * **4. Permutation Invariance:** Degradations applied in arbitrary undisclosed sequences.
- * **5. Dataset Partitioning:** Deterministic 80/20 Train/Validation split (seed=42) strictly isolating 640 unseen validation patterns.

3. End-to-End System Architecture Pipeline

Hierarchical Feature Encoding, Global Residual Learning & Sub-Pixel Upsampling

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- * **Input Stream:** Single-channel float32 raw degraded image (B, 1, 128, 128) ingested without heuristic pre-filtering.
- * **Global Residual Pathway:** Continuous bicubic upsampling provides base low-frequency structure (B, 1, 256, 256).
- * **Deep Feature Extractor:** 4-stage NAFNet-SR encoder-decoder architecture with skip-connections (29.33M parameters).
- * **Sub-Pixel Upsampling Head:** PixelShuffle(2) reconstructs fine sub-micron semiconductor edges and vias.
- * **Post-Processing & Output Contract:** Residual addition (Base + Res) followed by `torch.clamp(x, 0.0, 1.0)` output tensor (B, 1, 256, 256).

4. Preprocessing & Data Augmentation Strategy

Preserving Physical Fidelity While Preventing Overfitting

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- * **No Disallowed Preprocessing:** No heuristic denoising filters applied prior to model to avoid irreversible defect blurring.
- * **Geometric Symmetries:** Random horizontal flips ($p=0.5$), vertical flips ($p=0.5$), and random orthogonal rotations ($k \times 90$ deg).
- * **Synthetic Degradation Augmentation:** On-the-fly multi-order degradation engine generating diverse noise levels during training.
- * **High-Speed RAM Caching:** Reloads float32 arrays into system memory for instantaneous GPU batch streaming.

5. NAFNet-SR Architecture & Design Rationale

Nonlinear Activation Free Super-Resolution (Chen et al. Adaptation)

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- * **Key Principle:** Eliminates computationally expensive non-linearities (GELU/ReLU) in favor of SimpleGate element-wise multiplication.
- * **SimpleGate (SG):** $x1, x2 = \text{split}(x) \rightarrow \text{Output} = x1 * x2$. Captures non-linear gating at zero FLOP overhead.
- * **Simplified Channel Attention (SCA):** Global pooling + 1×1 Conv captures cross-channel wafer feature correlation.
- * **Depthwise Separable Convolutions:** 3×3 DWConv preserves ultra-fine line edge details while maintaining low parameter footprint.
- * **LayerNorm2d:** Exact channel-wise normalization per feature map preventing numerical divergence on flat wafer regions.

6. Composite Loss Formulation & Training Setup

Balancing Pixel Fidelity, Structural Similarity, Perceptual Quality & Frequency Spectrum

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- * **Total Loss:** $L_{\text{total}} = 1.0 * L_{\text{Charbonnier}} + 0.5 * L_{\text{SSIM}} + 0.1 * L_{\text{FFT}} + 0.02 * L_{\text{LPIPS}}$
- * **1. Charbonnier Loss (1.0):** Robust differentiable L1 loss avoiding oversmoothing in wafer defect boundaries.
- * **2. SSIM Structural Loss (0.5):** Preserves luminance, contrast, and topological structure across circuit traces.
- * **3. 2D-FFT Loss (0.1):** Enforces Fourier frequency spectrum consistency, recovering sharp repeating wafer pitch frequencies.
- * **4. LPIPS Loss (0.02):** AlexNet perceptual deep feature distance preventing blurry hallucination.
- * **Optimizer:** AdamW (lr=5e-4, min_lr=1e-6) with Cosine Annealing scheduler in pure FP32 / TF32 mode.

7. Sanity Check & Karpathy 2-Pair Overfit Test

Rigorous Empirical Verification of Model Capacity Prior to Full Convergence

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- * **Sanity Protocol:** Isolate exactly 2 degraded/GT image pairs and train without regularization.
- * **Overfit Verification Criterion:** Require Total Loss $\rightarrow 0.0$ and Peak PSNR > 40.0 dB.
- * **Empirical Result:** Model achieved PSNR 40.23 dB (> 40 dB target) in 80 iterations (Loss: 0.0018).
- * **Conclusion:** Proves gradient flow validity, loss formulation correctness, and zero structural bottlenecks in NAFNet-SR.

8. Validation Benchmark Results (Mean $\pm$ Std)

Rigorous Quantitative Performance on 80/20 Validation Split (N=640)

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- * **Bicubic Baseline:** PSNR: 23.01 ± 3.65 dB | SSIM: 0.5286 ± 0.1950 | LPIPS: 0.4428 ± 0.1618
- * **Classical U-Net (Baseline Repo):** PSNR: 27.17 ± 4.30 dB | SSIM: 0.7121 ± 0.1472 | LPIPS: 0.2600 ± 0.1100 (1.93M params)
- * **NAFNet-SR (Our Model):** PSNR: 28.16 ± 5.02 dB | SSIM: 0.7661 ± 0.1571 | LPIPS: 0.2277 ± 0.1222 (29.33M params)
- * **Net Delta vs Bicubic:** +5.15 dB PSNR Gain | +0.2375 SSIM Structural Gain | -48.6% Perceptual Distortion Drop
- * **Comparison:** Outperforms classical U-Net across all pixel, structural, and perceptual metrics.

9. Runtime Performance & Production Latency

Benchmarked on NVIDIA GeForce RTX 3050 Laptop GPU (4GB VRAM)

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- * **Single-Image Latency (Batch=1)** 39.10 $\pm$ 8.41 ms (25.6 FPS) on RTX 3050 Laptop GPU.
- * **Batched Effective Latency (Batch=8)**: 10.97 ms per image (91.2 FPS throughput).
- * **Peak Memory Footprint** 284.0 MB VRAM peak, easily deployable on memory-constrained inspection workstations.
- * **Hardware Context** RTX 3050 (2048 CUDA cores) is lower-tier compared to RTX 4050 (2560 cores), confirming efficient architecture.

10. Visual Analysis, Edge Sharpness & Limitations

Qualitative Restorations Across Noisy Wafer Patterns & Defect Boundary Analysis

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- * **Visual Fidelity:** Successfully removes dense speckle grain while preserving 1-pixel wide wafer interconnect lines.
- * **Frequency Recovery** 2D FFT loss recovers sharp periodic pitch lines without introducing hallucinated artifacts.
- * **Failure Case Analysis:** In extreme localized sensor saturation (>4 sigma noise), slight contrast suppression occurs.
- * **Mitigation:** Global residual bicubic base ensures baseline structural fidelity is never lost.

11. Conclusion, Repository & Compliance Summary

Phase 1 Submission Summary — KLA Hackathon 2026

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- * **GitHub Repository:** <https://github.com/Ad1thh/KLA-Semicon-AI-Restoration> (Public & Complete)
- * **Inference CLI Contract:** `python inference.py --input_dir <path> --output_dir <path>` (Strictly compliant)
- * **Reproducible Weights:** Tracked directly in Git repository (weights/nafnet_sr_best.pt, 56.15 MB).
- * **Deliverables:** Presentation (.pptx & .pdf), requirements.txt, 6 visual triplets, evaluation scripts.